

Daily Practice Problem (DPP)

Topic: Propositions, Implication, Biconditional, Truth tables, Predicates and Quantifiers

Name: _____

1. Match each logical connective with its meaning:

Connective	Meaning
1. $p \wedge q$	A. True unless p is true and q is false
2. $p \vee q$	B. True only for both have the same truth value
3. $p \rightarrow q$	C. True if at least one is true
4. $p \leftrightarrow q$	D. True if exactly one is true
5. $p \oplus q$	E. True only if both are true

2. Which of these are propositions? For those that are, state if they are true (T) or false (F).

- (a) $2 + 3 = 5$
- (b) $x + 1 = 3$
- (c) Close the window.
- (d) Every even integer greater than 2 can be expressed as the sum of two primes.
- (e) What time is it?
- (f) 7 is an odd number.

3. Let p , q , and r be the propositions

- p : You have the flu.
- q : You miss the final examination.
- r : You pass the course.

Express each of these propositions as an English sentence.

- (a) $p \rightarrow q$
- (b) $\neg q \leftrightarrow r$
- (c) $q \rightarrow \neg r$
- (d) $p \vee q \vee r$
- (e) $(p \rightarrow \neg r) \vee (q \rightarrow \neg r)$
- (f) $(p \wedge q) \vee (\neg q \wedge r)$

4. Construct a truth table for each of these compound propositions.

- (a) $p \rightarrow \neg p$
- (b) $p \leftrightarrow \neg p$
- (c) $p \oplus (p \vee q)$
- (d) $(p \wedge q) \rightarrow (p \vee q)$
- (e) $(q \rightarrow \neg p) \leftrightarrow (p \leftrightarrow q)$
- (f) $(p \leftrightarrow q) \oplus (p \leftrightarrow \neg q)$

5. Complete the missing values in the truth table for $(p \rightarrow q) \wedge (\neg q \rightarrow \neg p)$.

p	q	$p \rightarrow q$	$\neg q$	$\neg p$	$\neg q \rightarrow \neg p$	$(p \rightarrow q) \wedge (\neg q \rightarrow \neg p)$
T	T	T	F	F	T	
T	F		T	F		
F	T					
F	F					

6. The binary operation $\square$ is defined as follows:

- $P \square Q$ truth table:

P	Q	$P \square Q$
T	T	T
T	F	T
F	T	F
F	F	T

Which one of the following is equivalent to $P \vee Q$?

- (a) $\neg Q \square \neg P$
- (b) $P \square \neg Q$
- (c) $P \square Q$
- (d) $\neg P \square Q$

7. Translate the following English sentence into propositional logic notation:

“You can borrow the rare book only if you are a faculty member and have a valid ID.”

- Let F be “You are a faculty member,”
- I be “You have a valid ID,” and
- B be “You can borrow the rare book.”

8. Which one of the following Boolean expressions is NOT a tautology?

- (a) $((a \rightarrow b) \wedge (b \rightarrow c)) \rightarrow (a \rightarrow c)$
- (b) $(a \leftrightarrow c) \rightarrow (\sim b \rightarrow (a \wedge c))$
- (c) $(a \wedge b \wedge c) \rightarrow (c \vee a)$
- (d) $a \rightarrow (b \rightarrow a)$

Connecting Logic With Computer Science

9. What are the truth values of the following propositions?

- (a) $2^2 + 3^2 = 4^2$
- (b) The number 202 is written 11010010 in binary.
- (c) After executing the following C code, the variable x has the value 1.

```
int x = 202;
while (x > 2) {
    x = x / 2;    /* Note: in C, a single slash denotes integer division. */
}                /* For example, the value of 7/2 is 3, not 3.5. */
```

10. In many programming languages, the Boolean values **True** and **False** are actually stored as the numerical values 1 and 0, respectively. In Python, for example, both `0 == False` and `1 == True` are True. Thus, despite appearances, we can do arithmetic on Boolean values! Furthermore, in many languages (including Python), anything that is not False (in other words, anything other than 0) acts like True for the purposes of conditionals; for example, `if 2 then X else Y` will run successfully and execute X.

Suppose that x and y are two Boolean variables in a programming language where True and False are 1 and 0. (That is, the values of x and y are both 0 or 1.) Each of the following code snippets includes a conditional statement based on an arithmetic expression using x and y . Rewrite the given condition using the standard notation of propositional logic.

- a. `if x * y ...`
- b. `if x + y ...`
- c. `if 2 - x - y ...`
- d. `if (x * (1 - y)) + ((1 - x) * y) ...`
- e. Assume that $\{p_1, \dots, p_n\}$ are all Boolean variables in Python — that is, their values are all 0 or 1. Write a Python conditional expressing the condition that at least 3 of $\{p_1, \dots, p_n\}$ are true.
- f Write a Python conditional expressing the condition that at least $n - 1$ of $\{p_1, \dots, p_n\}$ are true.

11. In addition to purely logical operations, computer circuitry has to be built to do simple arithmetic very quickly. Consider a number

$$x \in \{0, \dots, 15\}$$

represented as a 4-bit binary number, and denote by x_0 the least-significant bit of x , by x_1 the next bit, and so forth. (Think of 0 as false and 1 as true.) Here you'll explore some pieces of using propositional logic and binary representation of integers to express arithmetic operations. (It's straightforward to convert your answers into circuits.) See Figure below.

x_3	x_2	x_1	x_0		x_3	x_2	x_1	x_0		x_3	x_2	x_1	x_0	
0	0	0	1		0	0	1	1		1	1	0	0	
				$0 + 0 + 0 + 1 = 1$					$0 + 0 + 2 + 1 = 3$					$8 + 4 + 0 + 0 = 12$

Fig (a). Representing small numbers with four Boolean variables (where 0 = False and 1 = True).

- Give a proposition over $\{x_0, x_1, x_2, x_3\}$ that expresses that x is greater than or equal to 8.
- Give a proposition over $\{x_0, x_1, x_2, x_3\}$ that expresses that x is evenly divisible by 4.
- Give a proposition over $\{x_0, x_1, x_2, x_3\}$ that expresses that x is evenly divisible by 5. (Hint: build a truth table.)
- Give a proposition over $\{x_0, x_1, x_2, x_3\}$ that expresses that x is evenly divisible by 9.
- Give a proposition over $\{x_0, x_1, x_2, x_3\}$ that expresses that x is an exact power of two.
- Suppose that we have two 4-bit input integers x and y , represented as in questions a–e. Give a proposition over the Boolean variables $\{x_0, x_1, x_2, x_3, y_0, y_1, y_2, y_3\}$ that expresses the condition that $x = y$.
- Give a proposition over $\{x_0, x_1, x_2, x_3, y_0, y_1, y_2, y_3\}$ that expresses the condition $x = 2 \cdot y$.
- Give a proposition over $\{x_0, x_1, x_2, x_3, y_0, y_1, y_2, y_3\}$ that expresses the condition $x^y = y^x$.

Challenge Problem

Under many operating systems, users can schedule a task to be run at a specified time in the future. In Unix-like operating systems, this type of scheduled job is called a *cron job*. (Greek: chron—“time.”) For example, a backup might run nightly at 2:00am, and a scratch drive might be emptied out weekly on Friday night at 11:50pm.

Let $T = \{1, 2, \dots, t_{\max}\}$ be a set of times (measured in minutes, let’s say), and let J be a set of jobs. Let *scheduledAt* be a predicate where *scheduledAt*(j, t) means “job j is scheduled at time t .” (Assume that jobs do not last more than one minute. The same job can be scheduled at multiple times.)

Formalize the following conditions using only standard quantifiers, arithmetic operators, logical connectives, and the *scheduledAt* predicate.

- (a) There is never more than one job scheduled at the same time.
 - (b) Every job is scheduled at least once.
 - (c) Job A is never run twice within 2 minutes.
 - (d) Job B is run at least three times.
 - (e) Job C is run at most twice.
 - (f) Job D is run sometime after the last time that Job E is run.
 - (g) Job F is run at least once between consecutive executions of Job G .
 - (h) Job H is run at most once between consecutive executions of Job I .
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Self-Evaluation

How confident do you feel after solving this practice sheet?

Low 1 2 3 4 5 High